

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 5 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTHM505 - PROJECT MANAGEMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Allocation of manpower, machinery, and materials is called:
A) Monitoring B) Planning
C) Resource Allocation D) Execution
 - 2 A project is selected after checking technical feasibility and financial viability. This activity belongs to:
A) Planning stage B) Scheduling stage
C) Project identification stage D) Execution stage
 - 3 When project budgeting, time planning, and method selection are performed together, it represents:
A) Identification B) Planning
C) Scheduling D) Execution
 - 4 The project manager reduces project duration by adding overtime only to critical activities. This technique is called:
A) Resource leveling B) Fast tracking
C) Crashing D) Updating
 - 5 The duration at which total project cost is minimum is called:
A) Normal duration B) Crash duration
C) Optimum duration D) Critical duration
 - 6 After 20 days of work, actual progress is entered and the network is recalculated. This process is called:
A) Crashing B) Fast tracking
C) Updating D) Resource allocation
 - 7 Central Limit Theorem in PERT is used to:
A) Estimate individual activity cost
B) Approximate total project duration distribution
C) Calculate slack of non-critical activities
D) Determine direct and indirect cost
 - 8 In PERT, the activity time that is most likely to occur is called:
A) Optimistic time B) Most likely time
C) Pessimistic time D) Expected time

- 9 Inflation refers to:
 A) Decrease in price level
 B) Increase in purchasing power
 C) Continuous rise in general price level
 D) Constant price level
- 10 When demand increases and supply remains constant, price will:
 A) Decrease
 B) Remain same
 C) Increase
 D) Become zero
- 11 CAPM stands for:
 A) Computer Assisted Process Management
 B) Computer Aided Project Management
 C) Control and Project Monitoring
 D) Construction Aided Project Model
- 12 Deming is famous for:
 A) 7 Quality Tools
 B) 14 Principles of Management
 C) Six Sigma
 D) ISO Standard

Q2. Solve the following.

- A) Explain Merits and Demerits of Gantt or Bar Chart. 6
- B) Explain the various steps involved in Project Management in detail. 6

Q3. Solve the following.

- A) Explain Resource Allocation and Resource Leveling by giving suitable example. 6
- B) Explain Network updating and also explain procedure for network updating. 6

Q4. Solve Any Two of the following.

- A) The table given below gives activities and their duration 6
 a. Draw Network. Calculate Project Duration and Highlight Critical Path
 b. Calculate EST, EFT, LST, LFT, Independent float and interfering float for the activities. Tabulate the result with sample calculation.

Activity	1-2	1-3	2-4	3-4	4-5
Duration (days)	2	3	4	5	6

- B) Differentiate between CPM and PERT. 6
- C) Define Pessimistic Time, Dummy activity and Optimistic Time. 6

Q5. Solve Any Two of the following.

- A) Explain Importance of Engineering Economics in Civil Engineering Field. 6
- B) Explain different types of costs used in engineering economics with suitable examples. 6
- C) A company wishes to invest in a new project. It has two alternatives A & B following data is available. Suggest which one is to be selected based on NPV and IRR. 6

Particulars	Project A	Project B
Initial Investment	10,50,000	12,75,000
Cash Inflows in Rs.		
Year 1	5,50,000	7,50,000
Year 2	7,00,000	5,00,000
Interest Rate	10%	10%

Q6. Solve Any Two of the following.

- A) Explain the importance of Total Quality Management (TQM) in project execution. 6
- B) Define Quality Control. Explain its objectives and importance in construction projects. 6
- C) Explain Break Even Analysis with neat Sketch. 6

***** End *****